

Tutorials – II

1. Consider an airfoil in a flow at standard sea level conditions with a freestream velocity of 50 m/s. At a given point on the airfoil, the pressure is $0.9 \times 10^5 \text{ Nm}^{-2}$. Calculate the velocity at this point.

Solution:

Using Bernoulli's equation:

$$p_{\infty} + \frac{1}{2}\rho V_{\infty}^2 = p + \frac{1}{2}\rho V^2$$

$$V = \sqrt{\frac{2(p_{\infty} - p)}{\rho} + V_{\infty}^2} = \sqrt{\frac{2(1.1 - 0.9) \times 10^5}{1.23} + (50)^2} = 142.8 \text{ m/s}$$

2. Consider the inviscid, incompressible flow of air along a streamline. The air density along the streamline is 1.225 kg/m^3 , which is the standard atmospheric density at sea level. At point 1 on the streamline, the pressure and velocity are 101314.6 N/m^2 and 3.05 m/s , respectively. Further downstream, at point 2 on the streamline, the velocity is 58 m/s . Calculate the pressure at point 2. What can you say about the relative change in pressure from point 1 to point 2 compared to the corresponding change in velocity?

Solution:

$$p_1 + \frac{1}{2}\rho V_1^2 = p_2 + \frac{1}{2}\rho V_2^2$$

$$p_2 = 99265.35 \text{ N/m}^2$$

As the flow velocity increases from 3.05 m/s to 58 m/s along the streamline, the pressure decreases from 101314.6 N/m^2 to 99265.35 N/m^2 . This is a factor of 19 increase in velocity, for only a factor of 0.02 decrease in pressure.

3. Consider a venturi with a throat-to-inlet area ratio of 0.8 mounted in a flow at standard sea level conditions. If the pressure difference between the inlet and the throat is 335.16 N/m^2 , calculate the velocity of the flow at the inlet.

Solution:

Using Bernoulli's equation and Continuity equation

For incompressible flow,

$$p_1 + \frac{1}{2}\rho V_1^2 = p_2 + \frac{1}{2}\rho V_2^2$$

$$\rho A_1 V_1 = \rho A_2 V_2 \Rightarrow A_1 V_1 = A_2 V_2 \Rightarrow \frac{V_2}{V_1} = \frac{A_1}{A_2}$$

Given $(A_2/A_1) = 0.8$

$$V_1 = \sqrt{\frac{2(p_1 - p_2)}{\rho \left[\left(\frac{A_1}{A_2} \right)^2 - 1 \right]}} = \sqrt{\frac{2(335.16)}{1.225 \left[(1/0.8)^2 - 1 \right]}} = 31.18 \text{ m/s}$$

- 4 Consider a model of an airplane mounted in a subsonic wind tunnel. The wind-tunnel nozzle has a 12-to-1 contraction ratio. The maximum lift coefficient of the airplane model is 1.3. The wing planform area of the model is 0.557 m^2 . The lift is measured with a mechanical balance that is rated at a maximum force of 454 kg; that is, if the lift of the airplane model exceeds 454 kg, the balance will be damaged. During a given test of this airplane model, the plan is to rotate the model through its whole range of angle of attack, including up to that for maximum C_L . Calculate the maximum pressure difference allowable between the wind tunnel settling chamber and the test section, assuming the standard sea level density in the test section.

Solution:

$$L_{\max} = 0.5 \times \rho_{\infty} \times V_{\infty}^2 \times S \times C_{L,\max} \Rightarrow V_{\infty} = \sqrt{\frac{2L_{\max}}{\rho_{\infty} S C_{L,\max}}} = \sqrt{\frac{2(454)}{1.225 \times 0.557 \times 1.3}} = 100.09 \text{ m/s}$$

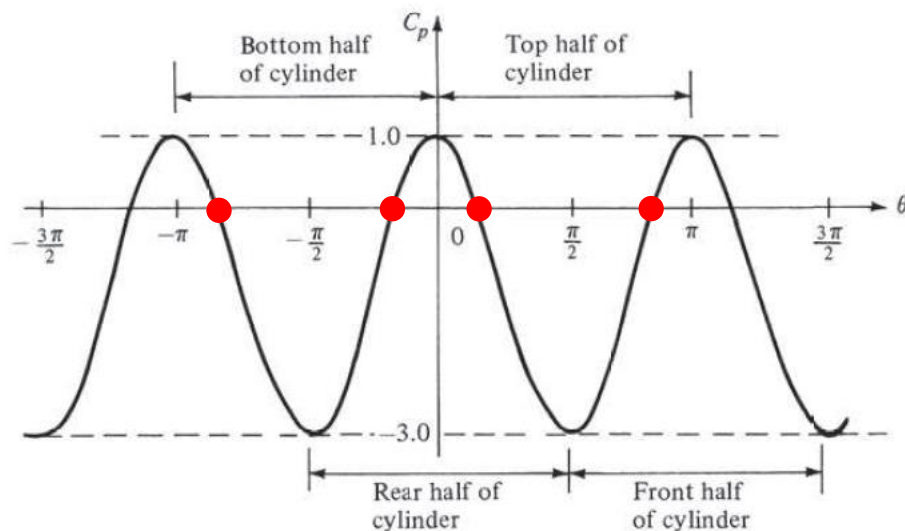
$$p_1 - p_2 = \frac{1}{2} \rho_{\infty} V_{\infty}^2 \left[1 - \left(\frac{A_2}{A_1} \right)^2 \right] = \frac{1}{2} (1.225) (100.09)^2 \left[1 - \left(\frac{1}{144} \right) \right] = 6094.64 \text{ Pa}$$

- 5 Consider an aircraft cruising at a standard altitude of 4 km (pressure = $6.166 \times 10^4 \text{ N/m}^2$ and density = 0.81935 kg/m^3). The pressure sensed by the Pitot tube on its right wing is $6.7 \times 10^4 \text{ N/m}^2$. At what velocity is the aircraft flying?

Solution:

$$V_1 = \sqrt{\frac{2(P_0 - P_1)}{\rho}} = \sqrt{\frac{2(6.7 - 6.166) \times 10^4}{0.81935}} = 114.2 \text{ m/s}$$

- 6 Consider the non-lifting flow over a circular cylinder. Mark the locations on the graph where the surface pressure equals the freestream static pressure.



When $p = p_\infty$, then $C_p = 0$

$$C_p = 1 - 4 \sin^2 \theta = 0$$

$$\sin \theta = \pm(1/2) \Rightarrow \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

- 7 Consider the lifting flow over a circular cylinder. The lift coefficient is 5. Find the location of the stagnation points and the points on the cylinder where the pressure equals freestream static pressure.

Solution:

$$\text{Lift coefficient, } C_l = \frac{\Gamma}{RU_\infty} = 5$$

The stagnation points are given by,

$$\theta = \arcsin\left(\frac{-\Gamma}{4\pi U_\infty R}\right) = \arcsin\left(-\frac{5}{4\pi}\right) = 203.4^\circ \text{ and } 336.6^\circ$$

To find the values of θ where $p = p_\infty$, then $C_p = 0$

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2$$

$$V = U = -2V_\infty \sin \theta - \frac{\Gamma}{2\pi R}$$

$$C_p = 1 - \left(\frac{-2V_\infty \sin \theta - \frac{\Gamma}{2\pi R}}{V_\infty}\right)^2 = 1 - 4 \sin^2 \theta - \frac{2\Gamma \sin \theta}{\pi R V_\infty} - \left(\frac{\Gamma}{2\pi R V_\infty}\right)^2$$

$$\text{Lift coefficient, } C_l = \frac{\Gamma}{RU_\infty} = 5$$

$$C_p = 1 - 4 \sin^2 \theta - \frac{10}{\pi} \sin \theta - \left(\frac{5}{2\pi}\right)^2 = 0.367 - 3.183 \sin \theta - 4 \sin^2 \theta$$

From the quadratic formula,

$$\sin \theta = \frac{3.183 \pm \sqrt{(3.183)^2 + 5.872}}{-8} = -0.897 \text{ or } 0.102$$

Hence,

$$\theta = 243.8^\circ \text{ and } 296.23^\circ$$

Also,

$$\theta = 5.85^\circ \text{ and } 174.1^\circ$$

Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.